Free Energy Analysis of Solvation with the Method of Energy Representation

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1 Solvation free energy

The chemical potential is the free-energy change corresponding to the insertion of a unit number of solute molecules in solution. It is actually the most fundamental quantity in the theory of solutions. Indeed, the free-energy change for a physical or chemical process of interest can be readily evaluated once the chemical potentials are known for the species present in the initial and final states of the process. Part of the chemical potential quantifying the solvent effect is the solvation free energy. It is the free-energy change for turning on the solute-solvent interaction. Since the solvent effect is the main focus of solution studies, it is of primary importance in statistical mechanics of solutions to establish a scheme to determine the solvation free energy of a solute in solution.

The purpose of this review is to describe schemes of computing the solvation free energy. The free energy in solution is notorious for its heavy computational demand. We approach this difficulty by combining the molecular simulation and statistical-mechanical theory of solutions. Since the target of solution chemistry expands and now includes quantum-classical coupled (QM/MM) system, nano-organized systems such micelle, lipid membrane, and protein, and environmentally benign reaction media such as ionic liquid and supercritical fluid, the theory of solutions also needs to be (re-)formulated to treat these frontline subjects. The goal of this review is to introduce a new theory of solutions which is amenable to diverse areas of applications. We first describe the standard scheme of free-energy calculation. We then present the concept of distribution function and the density-functional theory connecting the distribution function and free energy. Finally, we formulate a new method of solutions, the method of energy representation, and show its application to molecular binding into micelle and lipid membrane.

2 Free-energy perturbation and thermodynamic integration methods

Let H_0 and H_1 be respectively the Hamiltonians at the initial and final states of a process in solution. When the solvation process is concerned, the initial and final states are typically the pure solvent and solution systems of interest. The corresponding free-energy change ΔF is given by

$$\exp(-\beta\Delta F) = \frac{\int d\Gamma \exp(-\beta H_1)}{\int d\Gamma \exp(-\beta H_0)} \tag{2-1}$$

where β is the inverse of the product of the Boltzmann constant k_B and the temperature T and Γ is the (collective) coordinate for the phase space. When the classical statistical mechanics is adopted and the Hamiltonian change between the initial and final states does not involve the kinetic part, Eq. (2-1) reduces to

$$\exp(-\beta\Delta F) = \frac{\int d\mathbf{X} \exp(-\beta U_1)}{\int d\mathbf{X} \exp(-\beta U_0)} \tag{2-2}$$

where $\mathbf{X}$ is the (collective) coordinate for the configuration and U_0 and U_1 are the potential energies of the system at the initial and final states, respectively. Equation (2-2) is the starting point of our development. It should be noted that Eq. (2-2) cannot be used when a quantum fluid is to be treated.

When U_1- U_0 is denoted by ΔU, Eq. (2-2) is rewritten as

$$\exp(-\beta\Delta F) = \langle \exp(-\beta\Delta U) \rangle_0 \tag{2-3}$$

where $< ... >_0$ is the ensemble average taken at the initial state whose configuration is sampled according to the potential energy U_0. Equation (2-3) shows that the free-energy change ΔF can be calculated, in principle, by performing only the simulation for the initial state and averaging the Boltzmann factor of the relevant energy change ΔU. Indeed, Eq. (2-3) is the basis of the particle insertion method for evaluating the solvation free energy (chemical potential) of a solute in solution [1-3]. In the particle insertion method, the pure solvent is simulated and the solute molecule of interest is inserted randomly into the pure solvent. In this insertion process, the configuration of the solvent molecules constituting the pure solvent system is unchanged. The solute-solvent interaction energy is calculated by inserting a solute molecule without changing the solvent configuration. Such a solute molecule is denoted as a test particle. The chemical potential is then obtained from

$$\exp(-\beta\Delta F) = \int d(\Delta U) \exp(-\beta\Delta U) f(\Delta U) \tag{2-4}$$

where $f(\Delta U)$ is the probability distribution function of ΔU in the pure solvent system.

The particle insertion method is convenient and fast since only the pure solvent configurations need to be prepared and the free energy is calculated from a one-step insertion process of the solute. As is well documented, however, the particle insertion method is applicable only for a small and weakly interacting solute [1-3]. When the

solute is large and/or interacts strongly with the solvent, the formally exact form of Eq. (2-4) exhibits a numerical difficulty. See Figure 1. When the solute is not small, it almost always overlaps with solvent molecules upon insertion. When the solute does not interact weakly with the solvent, the attractive solute-solvent effect needs to be properly taken into account. The problem is that $f(\Delta U)$ is well sampled only toward large (repulsive) value of ΔU. Although the Boltzmann factor exp $(-\beta\Delta U)$ increases steeply toward small (attractive) ΔU, the small ΔU region, which makes a significant contribution in Eq. (2-4), is usually ill sampled in the computation of $f(\Delta U)$. Thus, Eq. (2-4) is not computationally useful and the particle insertion method cannot be used for most of "interesting" systems. Actually, the calculation of the average of the exponential of the energy change is often prohibitive unless the energy change is small in magnitude. When large values (exp $(-\beta\Delta U)$ in the present case) is not well sampled, the numerical averaging is ill behaved. This is a common difficulty in numerical statistical analysis.

The standard and often used methods to circumvent the difficulty associated with the form of Eq. (2-4) are the free-energy perturbation and thermodynamic integration methods [1-3]. These methods are generally applicable to free-energy evaluation. In this review, we restrict our development to the solvation process; the initial state is the pure solvent and the final state is the solution system of interest.

The free-energy perturbation method utilizes the intermediate states connecting the initial and final states of the process of interest. Let V_i (i = 0, …, n) be a sequence of potential energies where the initial and final ones V_0 and V_n are taken to the potential functions U_0 and U_1 for the initial and final states, respectively. For an arbitrary (set of) V_i, Eq. (2-2) can be expressed as

$$\exp(-\beta\Delta F) = \prod_{i=0}^{n-1} \langle \exp(-\beta(V_{i+1} - V_i)) \rangle_i \tag{2-5}$$

where $< \ldots >_i$ is the ensemble average taken with respect to the potential function V_i. Equation (2-5) shows that ΔF is given as the sum of the free-energy change accompanying the energy change from V_i to V_{i+1} (i = 0, …, n-1). The states corresponding to V_i (i = 1, …, N-1) are called intermediate states. The free energy is a state function and does not depend on the choice of the intermediate states in principle. From the computational viewpoint, the point is to "select" the set of V_i so that the change from V_i to V_{i+1} is "small" in magnitude. When V_i and V_{i+1} are "similar" and the energy change is small, the difficulty encountered in the particle insertion method can be circumvented and the calculation of the free-energy change becomes feasible. The drawback is that a number of intermediate states need to be prepared and that the computational cost is enhanced accordingly.

In the thermodynamic integration method, the intermediate states are introduced with respect to the coupling parameter λ ($0 \le \lambda \le 1$). The potential function at the coupling parameter of λ is denoted as U_λ and satisfies $U_\lambda = U_0$ and $U_\lambda = U_1$ at the initial and final states (λ = 0 and 1), respectively. The intermediate states correspond to $0 < \lambda < 1$. The form of averaging-the-exponential is then avoided by rewriting Eq. (2-2) as

$$\Delta F = \int_0^1 d\lambda \left\langle \frac{\partial U_\lambda}{\partial \lambda} \right\rangle_\lambda \tag{2-6}$$

where $< ... >_\lambda$ is the ensemble average when the potential energy is U_λ. As is the case of the free-energy perturbation method, ΔF value calculated by Eq. (2-6) is independent of the choice of the intermediate states in principle. The integrand of Eq. (2-6) is a preferable average from the computational viewpoint. The exponential average is not involved any more. In practice, the integral of Eq. (2-6) is replaced by a discretized sum and a finite number of intermediate states are to be treated explicitly. Since a systematic error is introduced by the discretization, a large number of intermediate states need to be prepared and the computational demand increases correspondingly.

In both the free-energy perturbation and thermodynamic integration methods, the key to the computational accuracy and efficiency is the choice of the intermediate states as a function of the coupling parameter λ. Note that the intermediate states adopted in the free-energy perturbation method can be considered a finite subset of the intermediate states introduced continuously over $0 < \lambda < 1$. A straightforward implementation of the intermediate states is possible by varying the system potential energy linearly. When the solvation is concerned and the solute-solvent interaction is expressed as the sum of Lennard-Jones and Coulombic terms, the linear variation is realized by the intermediate solute-solvent interaction given by

$$\lambda \sum_{i,j} \left(4\varepsilon_{ij} \left\{ \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right\} + \frac{q_i q_j}{r_{ij}} \right) \tag{2-7}$$

where i and j refer to solute and solvent interaction sites, respectively. The first term in the sum expresses the Lennard-Jones interaction at the distance r_{ij} between the solute and solvent sites, and ε_i and σ_i are the energy and length parameters, respectively. The second term in the sum corresponds to the Coulombic interaction, and q_i and q_j are the charges on the solute and solvent sites, respectively. The linear scaling of the solute-solvent interaction with Eq. (2-7) is often ill-behaved numerically around $\lambda = 0$. This is related to the appearance of $r = 0$ singularity at $\lambda = 0$. To alleviate the problem, a nonlinear scaling, which is obtained by replacing λ with λ^n ($n \ge 2$) in Eq. (2-7), can be

used [4,5]. The calculation becomes more stable near $\lambda = 0$. The $r = 0$ singularity is still present in the nonlinear scaling, however, and careful analysis is necessary to determine the λ values actually sampled. A more promising choice of the intermediate states is provided by

$$4\lambda\varepsilon\left(\left(\frac{\sigma^2}{r^2+(1-\lambda)\delta}\right)^6-\left(\frac{\sigma^2}{r^2+(1-\lambda)\delta}\right)^3\right) \tag{2-8}$$

or similar expressions for the Lennard-Jones part [5,6]. The $r = 0$ singularity is then removed and the free-energy calculation on the basis of Eq. (2-8) is numerically stable.

Although the free-energy perturbation and thermodynamic integration methods are exact in principle under a given set of potential functions, they are not free from systematic errors in practice. The systematic error most often encountered in the free-energy perturbation method is the non-coincidence of the free-energy changes ΔF calculated from the forward variation of the coupling parameter λ from 0 to 1 and the backward variation from 1 to 0. The common practice is to average the ΔF from the forward and backward calculations. It is pointed out, however, that the simple averaging is itself a source of systematic error [7]. To achieve the accuracy, the use of Bennett's weighting function is recommended [7,8]. In the thermodynamic integration method, a systematic error is inevitable when the integral over λ in Eq. (2-6) is discretized. A careful examination of discretization is necessary, especially when the integrand of Eq. (2-6) exhibits a non-monotonic dependence on λ and/or varies steeply over some range of λ.

3 Distribution function theory

A molecular picture of solutions is established through distribution (correlation) functions. Correspondingly, a molecular description of the solvation free energy can be implemented by formulating a functional which expresses the solvation free energy in terms only of distribution functions in the solution and pure solvent systems of interest. An approximate functional needs to be constructed in practice, however, since the exact functional involves an infinite series of many-body distribution functions [9]. The theories introduced in Secs. 5 and 6 are formulated to provide the solvation free energy with simple distribution functions in closed form. In this section, a general description of distribution function is provided.

The system of our interest is a dilute solution containing a single solute molecule. Even when the solute concentration is finite, our development is valid by viewing one of the solute molecules as the "solute" and the others as part of mixed solvent. To

describe completely the solute-solvent configuration, the position and orientation need to be specified simultaneously. The complete set of the position and orientation is called the full coordinate and is denoted collectively as **x** for the solvent molecule. If the solvent molecule is flexible, its intramolecular degrees of freedom are also incorporated into **x**. Similarly, the full coordinate of the solute molecule is denoted as ψ. In the full coordinate representation, the instantaneous distribution $\hat{\rho}^f$ at an arbitrarily given snapshot configuration is introduced as

$$\hat{\rho}^f(\psi,\mathbf{x}) = \delta(\psi - \hat{\psi})\sum_i \delta(\mathbf{x} - \hat{\mathbf{x}}_i) \tag{3-1}$$

where $\hat{\mathbf{x}}_i$ and $\hat{\psi}$ are the full coordinate of the ith solvent molecule and the solute molecule at the snapshot configuration, respectively, and the sum is taken over all the solvent molecules. The superscript *f* is attached to emphasize that Eq. (3-1) is in the full coordinate representation. The distribution functions are generated from the averages of products of $\hat{\rho}^f$ in the system of interest.

When the distribution function is generated from $\hat{\rho}^f$, a multidimensional description is inevitable. Indeed, the full coordinate **x** is 6-dimensional (5 for linear molecule) for rigid species and involves more for flexible species. To implement the full coordinate representation, the expansion in terms of spherical harmonics can be employed [1]. However, the calculation of multidimensional distribution functions is often slow in molecular simulation and the numerical realization is not straightforward. When the coordinate has too "fine" information content, the corresponding distribution function needs additional methodology for handling. Too much information is not desirable both from the computational viewpoint (large memory and slow convergence) and the conceptual viewpoint (unclear perception in mind).

It is then useful to reduce the information content by introducing a "projected" coordinate. With projection, some information of (ψ,**x**) is retained, while the others are disregarded. When the projection is implemented with respect to a function $P(\psi,\mathbf{x})$, the corresponding distribution functions are generated from the instantaneous distribution given by

$$\sum_i \delta(p - P(\psi,\mathbf{x}_i)) \tag{3-2}$$

where *p* is the value of $P(\psi,\mathbf{x})$ and serves as the coordinate for the distribution function.

A typical choice of $P(\psi,\mathbf{x})$ is the radial distance between the atomic sites (interaction sites) of the molecule. When a pair of atomic sites in the solute and solvent molecules is picked up, the histogram of its radial distance is averaged with an appropriate normalization to give the site-site radial distribution function. For example, when the

solute and solvent is both H_2O (when one of the molecules in pure water is viewed as the "solute" and the others as the "solvent"), the O-O, O-H, and H-H radial distribution functions are generated by the projections onto O-O, O-H, and H-H distances, respectively. It should be noted that the radial distribution functions do not represent a simultaneous distribution of a set of site-site radial distances. In the case of water, the O-O radial distribution function specifies only the O-O distance and the other distance information such as those for O-H and H-H is disregarded. Similarly, the O-H radial distribution function does not contain explicit information about the O-O and H-H distances. In Figure 2, we show the O-O and O-H radial distribution functions of water at 1 g/cm^3 and 25 °C. It is seen from the O-O radial distribution function that the neighboring water molecules stay in the distance of ~2.8 Å. The O-H radial distribution function shows that the intermolecular hydrogen-bonding distance is ~2 Å. The second peak of the O-O radial distribution function is characteristic of water. In simple liquid, the second peak appears at about twice the distance for the first peak. In water, the second-peak position is ~1.6 times of the first-peak position. This provides a view that the ice-like structure persists even in liquid water.

Another useful distribution function is for the interaction pair energy. An example is shown in Figure 3 for water at 1 g/cm^3 and 25 °C. The peak at ~-6 kcal/mol corresponds to the intermolecular hydrogen bonding of water in the liquid state. When a pair of water molecules are far apart, it contributes to the distribution at $\varepsilon = \sim 0$. Since a large number of pairs with $\varepsilon = \sim 0$ are present, Figure 3 shows a high peak at $\varepsilon = \sim 0$. In the high-energy regime, the distribution function vanishes. This reflects the excluded volume effect and is consistent with the fact the radial distribution functions vanish at short distances.

Of course, the choice of the projecting function $P(\psi,\mathbf{x})$ of Eq. (3-2) is not unique. The choice depends on the purpose. For example, when the angle of the hydrogen bonding is of interest, it is most useful to adopt the hydrogen-bonding angle itself as $P(\psi,\mathbf{x})$ [10]. In general, no *a priori* criterion is present for preferable projection. The desirable form of projection can be based only upon the target quantity to be investigated.

4 Kirkwood charging formula and density-functional theory

A connection between the solvation free energy $\Delta\mu$ and distribution function is provided by the Kirkwood charging formula [1-3]. In the present section, we describe the Kirkwood charging formula for a pairwise additive solute-solvent interaction. The Kirkwood formula is then transformed into a density-functional form through partial

integration. We briefly describe the density-functional theory in the full coordinate representation.

The target quantity of the development is the solvation free energy. The solvation free energy $\Delta\mu$ is the free-energy change corresponding to the gradual insertion process of the solute molecule. In the classical (non-quantum) treatment of $\Delta\mu$, only the contribution from the potential energy is involved and the ideal (kinetic) contribution is excluded. Let ψ and $\mathbf{x}$ be the full coordinates of the solute and solvent molecules, respectively. We suppose that the solute-solvent intermolecular interaction of interest is pairwise additive and is written as $v(\psi,\mathbf{x})$. When the one-body energy of the solute including the intramolecular component is $\Psi(\psi)$ and the total solvent-solvent interaction energy is $U(\mathbf{X})$, $\Delta\mu$ is expressed as

$$\exp(-\beta\Delta\mu)=\frac{\int d\psi d\mathbf{X}\exp\left(-\beta\left\{\Psi(\psi)+\sum_i v(\psi,\mathbf{x}_i)+U(\mathbf{X})\right\}\right)}{\int d\psi d\mathbf{X}\exp\left(-\beta\left\{\Psi(\psi)+U(\mathbf{X})\right\}\right)} \tag{4-1}$$

where $\mathbf{X}$ represents the solvent configuration collectively and β is the inverse of the product of the Boltzmann constant k_B and the temperature T. A restriction of attention to a certain set of solute intramolecular state can be made simply by the corresponding alteration of the domain of integration over ψ.

To formulate the Kirkwood charging formula, we introduce a set of intermediate states involving pairwise additive solute-solvent interaction. Let λ be the coupling parameter identifying the state and $u_\lambda(\psi,\mathbf{x})$ be the solute-solvent interaction potential at the coupling parameter λ. When $\lambda = 0$, the system is the pure solvent system and $u_0(\psi,\mathbf{x}) = 0$ (no solute-solvent interaction). When $\lambda = 1$, the solute interacts with the solvent at full coupling and $u_1(\psi,\mathbf{x}) = v(\psi,\mathbf{x})$. The form of $u_\lambda(\psi,\mathbf{x})$ at $0<\lambda<1$ is arbitrary. The Kirkwood charging formula is an integration over the coupling parameter and is expressed as

$$\Delta\mu=\int_0^1 d\lambda\int d\psi d\mathbf{x}\frac{\partial u_\lambda(\psi,\mathbf{x})}{\partial\lambda}\rho^f(\psi,\mathbf{x};u_\lambda) \tag{4-2}$$

where $\rho^f(\psi,\mathbf{x};u_\lambda)$ is the ensemble average of Eq. (3-1) in the presence of the solute-solvent interaction u_λ. The superscript f means that the function is represented over the full coordinate $(\psi,\mathbf{x})$.

The solvation free energy can be evaluated using the Kirkwood charging formula [1-3]. It introduces a set of intermediate states connecting the initial and final states of the gradual insertion process of the solute; the initial and final states correspond to the pure solvent system without the solute and the solution system of interest, respectively.

In principle, the Kirkwood charging formula provides the "exact" solvation free energy under a given set of potential functions. Its implementation is expensive in practice, however, since a number of intermediate states need to be introduced. For the purpose of analyzing the solvation free energy on the molecular level, furthermore, it is necessary to express the solvation free energy in terms only of distribution functions in the solution and pure solvent systems of interest. Within the framework of the Kirkwood charging formula, the intermediate states are actually arbitrary and are employed for the convenience of formulation and computation. They are not of physical significance since the free energy is a state function.

A molecular description of the solvation free energy can be implemented by formulating a functional which expresses the solvation free energy in terms of distribution functions in the solution and pure solvent systems of interest. Distribution functions in the other systems should not be involved in the functional in order to assure that the free energy is given as a state function. The exact functional is not useful, however, since it is an infinite series of many-body distribution functions [9]. In practice, an approximate but accurate functional needs to be constructed which is expressed with few-body distribution functions in closed form. When such a functional is formulated and the distribution functions constituting the approximate functional are readily obtained by computer simulation, the solvation free energy can be determined and analyzed with reasonable computational load in terms of exact microscopic information of the systems of interest.

It is a statistical-mechanical theory of solutions to express the solvation free energy as a functional of distribution functions. Traditionally, the theory of solutions is formulated with a diagrammatic approach [1], in which an approximation is provided through a two-step procedure. In the first step, the free energy and/or distribution function is expanded with respect to the solute-solvent interaction potential function or its related function as an infinite, perturbation series. In the second step, a renormalization scheme is applied; a set of functions are defined through partial summation of the series and are employed for substitution to make the infinite series more tractable. An approximation is typically introduced simply by neglecting non-tractable diagrams.

We adopt an alternative route to the distribution function theory. The approach is based upon the density-functional theory. In this approach, the change of variables is conducted through Legendre transform from the solute-solvent interaction potential function to the solute-solvent distribution function or the solvent density around the solute. The (solvation) free energy is then expressed approximately by expanding the

corresponding Legendre-transformed function with respect to the distribution function to some low order.

The starting point of the density-functional treatment is the Kirkwood charging formula. The partial integration of the Kirkwood formula given by Eq. (4-2) provides

$$\begin{aligned}\Delta\mu &= \int d\psi d\mathbf{x} v(\psi,\mathbf{x})\rho^f(\psi,\mathbf{x};v) - \int_0^1 d\lambda \int d\psi d\mathbf{x} u_\lambda(\psi,\mathbf{x})\frac{\partial\rho^f(\psi,\mathbf{x};u_\lambda)}{\partial\lambda} \\ &\equiv \int d\psi d\mathbf{x} v(\psi,\mathbf{x})\rho^f(\psi,\mathbf{x}) - F^f\left[\rho^f(\psi,\mathbf{x})\right]\end{aligned} \tag{4-3}$$

where the density-functional F^f is defined in the second line of Eq. (4-3) and $\rho^f(\psi,\mathbf{x};v)$ at the potential v of interest is simply denotes as $\rho^f(\psi,\mathbf{x})$. $\Delta\mu$ and F^f are related to each other with Legendre transform since the map is proved to be one-to-one from the solute-solvent interaction potential to the distribution function [1]. An approximation can be devised by introducing the indirect part ω^f of the potential of mean force as

$$\rho_\lambda^f(\psi,\mathbf{x}) = \rho_0^f(\psi,\mathbf{x})\exp\left(-\beta\left(u(\psi,\mathbf{x};\rho_\lambda^f) + \omega^f(\psi,\mathbf{x};\rho_\lambda^f)\right)\right) \tag{4-4}$$

where $\rho^f_\lambda(\psi,\mathbf{x})$ is the distribution function at the coupling parameter λ. In Eq. (4-4), the dependence is written in terms of the distribution function ρ^f_λ, instead of the potential u_λ. This is possible due to the property of one-to-one correspondence. When the solvent-solvent correlation is absent (low-density limit), ω^f is zero. In other words, all the "complicated" solvent-solvent correlations are put into ω. Equations (4-3) and (4-4) lead exactly to

$$F^f\left[\rho^f(\psi,\mathbf{x})\right] = k_B T \int d\psi d\mathbf{x} \begin{bmatrix} \left(\rho^f(\psi,\mathbf{x}) - \rho_0^f(\psi,\mathbf{x})\right) - \rho^f(\psi,\mathbf{x})\log\left(\dfrac{\rho^f(\psi,\mathbf{x})}{\rho_0^f(\psi,\mathbf{x})}\right) \\ -\beta\left(\rho^f(\psi,\mathbf{x}) - \rho_0^f(\psi,\mathbf{x})\right)\int_0^1 d\lambda\,\omega^f(\psi,\mathbf{x};\rho_\lambda^f) \end{bmatrix} \tag{4-5}$$

where u_λ is taken so that ρ^f_λ varies linearly against λ, and $\rho^f(\psi,\mathbf{x})$ and $\rho^f_0(\psi,\mathbf{x})$ denote the distribution functions at the coupling parameter $\lambda = 0$ and 1, respectively. Equation (4-5) is exact, and an approximation is introduced to the λ integral of ω^f. When ω^f is taken to vary linearly with λ, the HNC (hypernetted-chain) approximation is obtained. When exp $(-\beta\omega^f)$-1 is set to be linear, it is the PY (Percus-Yevick) approximation.

The above is the brief introduction to the density-functional theory of solutions. The mathematical development is quite straightforward. The numerical implementation is difficult, however, in the full coordinate representation. As noted in Sec. 3, the full coordinate is multidimensional; the solute-solvent distribution is a function over high-dimensional configuration space and cannot be implemented in practice. To

overcome the problem of dimensionality, it is necessary to introduce a projected coordinate. In Sec. 6, we introduce the energy representation and formulate the density-functional theory in the energy representation.

5 Site-site radial distribution function and reference interaction site model

The method of reference interaction site model (RISM) is based on the site-site radial distribution functions. It introduces the "direct correlation functions" through the inverses of the correlation matrices for the site-site distance and formulates an approximate set of integral equation for the site-site radial distribution functions by adopting "closure" relationships between the radial distribution functions and direct correlation functions [1,11-14]. Compared to the molecular simulation method, the method of integral equation is much faster. The speed is achieved by restricting the attention only to the radial distribution functions and adopting approximate closures. Furthermore, the solvation free energy is expressed in closed form for some types of closure relationships [13,14]. In this case, no reference to the intermediate states of the solute insertion process is required and the solvation free energy can be evaluated directly from the radial distribution functions obtained from the integral equation. When a closed-form functional for the solvation free energy is given in terms of distribution functions, the functional not only provides an efficient route of computation, but also sets a basis for the molecular understanding with respect to the distribution functions.

A basic quantity in the RISM approach is the intermolecular site-site radial distribution function $g_{\alpha\gamma}(r)$, where α and γ denotes interaction sites and r is the radial distance between the sites α and γ. An example is provided in Figure 2. Another basic quantity in RISM is the intramolecular correlation function $\omega_{\alpha\gamma}(r)$. This carries information of the molecular structure, and its Fourier-transformed form is expressed over the reciprocal coordinate k as

$$\omega_{\alpha\gamma}(k) = \delta_{\alpha\gamma} + (1-\delta_{\alpha\gamma})\sin(kL_{\alpha\gamma})/kL_{\alpha\gamma} \tag{5-1}$$

where $L_{\alpha\gamma}$ is the intramolecular distance between the sites α and γ. It should be noted that $g_{\alpha\gamma}(r)$ and $\omega_{\alpha\gamma}(r)$ do not represent a simultaneous distribution of a set of site-site radial distances. Instead, they are a set of independent distributions. $g_{\alpha\gamma}(r)$ is provided by projecting the full-coordinate distribution function generated from Eq. (3-9) onto the site-site radial distance between α and γ with appropriate normalization. Similarly, $\omega_{\alpha\gamma}(r)$ is a projection of the one-body distribution function in the full coordinate representation onto the radial distance between a pair of sites α and γ.

The RISM integral equation is formulated in terms of the site-site total correlation

function $h_{\alpha\gamma}(r)$ and the site-site direct correlation function $c_{\alpha\gamma}(r)$. $h_{\alpha\gamma}(r)$ is defined from the site-site radial distribution function $g_{\alpha\gamma}(r)$ as

$$h_{\alpha\gamma}(r) = g_{\alpha\gamma}(r) - 1 \tag{5-2}$$

and $c_{\alpha\gamma}(r)$ is introduced through the site-site Ornstein-Zernike (SSOZ, also called RISM) equation written as

$$h_{\alpha\gamma}(|\mathbf{r}-\mathbf{R}|) = \sum_{\beta,\kappa}\int d\mathbf{x}d\mathbf{y}\omega_{\alpha\beta}(|\mathbf{r}-\mathbf{x}|)c_{\beta\kappa}(|\mathbf{x}-\mathbf{y}|)[\omega_{\kappa\gamma}(|\mathbf{y}-\mathbf{R}|)+\rho h_{\kappa\gamma}(|\mathbf{y}-\mathbf{R}|)] \tag{5-3}$$

where the system is supposed to be a one-component, homogeneous fluid for simplicity and ρ is the density of the fluid. Equation (5-3) was first derived from the Ornstein-Zernike equation in the full coordinate representation by adopting the assumption that the direct correlation function in the full coordinate representation is given as a sum of the site-site direct correlation functions. Instead, Eq. (5-3) can be viewed as a definition of the site-site direct correlation functions [15,16]. In any case, another set of equations, called "closure" relationship, is necessary to obtain $h_{\alpha\gamma}(r)$ and $c_{\alpha\gamma}(r)$ with Eq. (5-3). In the original formulation [11], the PY (Percus-Yevick)-type form was adopted as

$$c_{\alpha\gamma}(r) = [h_{\alpha\gamma}(r)+1][1-\exp(\beta v_{\alpha\gamma}(r))] \tag{5-4}$$

where $v_{\alpha\gamma}(r)$ is the interaction potential between the sites α and γ and β is the inverse of the product of the Boltzmann constant and the temperature. In the extended version [12], the HNC (hypernetted-chain)-type approximation was employed with

$$h_{\alpha\gamma}(r)+1 = \exp(-\beta v_{\alpha\gamma}(r)+h_{\alpha\gamma}(r)-c_{\alpha\gamma}(r)) \tag{5-5}$$

Equation (5-3) with Eq. (5-4) or (5-5) constitutes a self-consistent set of integral equations for $h_{\alpha\gamma}(r)$ and $c_{\alpha\gamma}(r)$. With the site-site potential $v_{\alpha\gamma}(r)$ and the intramolecular correlation function $\omega_{\alpha\gamma}(r)$ as inputs, the correlation functions $h_{\alpha\gamma}(r)$ and $c_{\alpha\gamma}(r)$ are given as outputs in the RISM treatment. Furthermore, the solvation free energy is expressed in closed form for certain types of closure relationships [13,14]. In this case, the integral equation provides not only the structural information $h_{\alpha\gamma}(r)$ but also the solvation free energy in an efficient manner. The RISM integral equation with PY-type or HNC-type closure can be solved with a variational procedure [11,12], though it is not of sound character from the standpoint of the density-functional theory [16].

A drawback is present, of course, in any approximate method of solutions. Under a given set of potential functions, the molecular simulation gives the exact distribution functions when it is done long enough. In contrast, since the closure relationship is approximate, the radial distribution function obtained from the integral equation method

is approximate. The solvation free energy calculated from the integral equation theory has two sources of errors. One is due to the approximate nature of the potential functions (force field), and the other comes from the approximation involved in the integral equation.

The drawbacks characteristic of RISM and its variants are related to the fact that they do not treat the whole molecule as a single unit and view a molecule as a collection of interaction sites. The method is thus applicable only when the potential function is of site-site form. Indeed, Eqs. (5-3)-(5-5) require a set of potentials in the site-site form as inputs. As a consequence, the electronic distribution cannot be treated in the cloud-like form as implemented in quantum theories, but needs to be contracted into a set of point charges. In addition, the integral equation is ill-behaved unless all of the interaction sites carry the repulsive core explicitly. For example, many of potential functions of H_2O do not assign a repulsive core at the H-site. This is simply because the repulsive core for the O-site is large enough that other molecules cannot come too close to the H-site. The repulsive core of the H-site is buried within the O-site core and is not necessary to be treated explicitly at the level of potential functions. In RISM, however, a core parameter needs to be assigned to the H-site, too. The core parameter actually dictates the resulting solvation free energy sensitively and acts as an adjustable parameter in the method. Another, related problem is so-called the "problem of auxiliary site". The solution to the RISM integral equations exhibits unphysical dependence on the presence of auxiliary sites which simply label points in a molecule and make no contribution to the intermolecular interaction. This type of difficulty is absent when the whole molecule is treated as a single unit. The difficulty arises when a molecule is treated as a collection of sites. In RISM, the correlation between a pair of sites is described at the two-body level for both the intramolecular and intermolecular ones. Since the sites in a molecule are tightly bound with one another, a partial incorporation of the intramolecular correlation is not desirable. This point is exemplified when the density is low. The RISM integral equations are not exact in the limit of zero solvent density and are not suitable to evaluate the solvation free energy in a low-density fluid. It is well-known in this instance, too, that the low-density limit is given exactly when the whole molecule is treated as a single unit in the full coordinate representation [1]. Finally, since the molecular structure is an input in the RISM approach, an additional scheme needs to be devised to deal with flexible molecules. Equations (5-3)-(5-5) take the intramolecular correlation functions $\omega_{\alpha\beta}(r)$ (or the intramolecular distance $L_{\alpha\beta}$) as inputs, and the variation of $\omega_{\alpha\beta}(r)$ in response to intermolecular interaction can be incorporated by formulating another set of (approximate) equations.

In the commonly used RISM approach, the solvation free energy is often expressed in closed form in terms of site-site radial distribution functions. An improvement of the approach may then be possible through combination with the molecular simulation; the radial distribution functions are exact under the used set of potential functions when they are calculated from the molecular simulation, instead of the integral equation. This line of approach is developed by Kast and Truong [17-19], and the computational efficiency is achieved compared to the free-energy perturbation and thermodynamic integration methods. In this approach, the site-site radial distribution functions are calculated from molecular simulations and are substituted into a RISM-type functional for the solvation free energy. The procedure suffers from numerical instability of several tens of kcal/mol, however. The source of the numerical trouble is the intramolecular correlation matrices in the range of small reciprocal vector (large distance). When the reciprocal vector approaches zero, the intramolecular correlation matrices becomes ill-conditioned and the direct correlation function exhibits a long-range decay of Coulombic functional form.

The above drawbacks of RISM and its variants are well documented since their first formulations [1,11,15,16]. They are all related to the point that a molecule is treated as a collection of sites. In the method of energy representation introduced in Sec. 3.6, each of the solute and solvent molecules is taken to be a single unit as a whole, and those drawbacks vanish.

6 Method of energy representation

In the method of energy representation, the projecting function $P(\psi,\mathbf{x})$ of Eq. (3-2) is taken to be the solute-solvent pair interaction energy. Figure 3 is an example of the distribution function in the energy representation. To introduce the energy representation, it is necessary to specify the solute-solvent interaction potential v of interest. Of course, v is a function of the solute configuration ψ and the solvent configuration $\mathbf{x}$. The instantaneous distribution $\hat{\rho}^e$ is defined as

$$\hat{\rho}^e(\varepsilon) = \sum_i \delta(v(\psi,\mathbf{x}_i) - \varepsilon) \tag{6-1}$$

where the sum is taken over the solvent molecules and a superscript e is attached to emphasize that a function is represented over the energy coordinate. Equation (6-1) is the histogram of the pair interaction energy between solute and solvent in a snapshot configuration of interest. The distribution functions in the energy representation are generated from the averages of products of $\hat{\rho}^e$ in the system of interest.

In the energy representation, the density-functional theory can be formulated by

restricting the set of solute-solvent interaction potentials $u_\lambda(\psi,\mathbf{x})$ to those which are constant over an equienergy surface of $v(\psi,\mathbf{x})$. In this case, when the value of $v(\psi,\mathbf{x})$ is denoted as ε, the intermediate states can be written as $u_\lambda(\varepsilon)$. At the end points ($\lambda = 0$ and 1), $u_0(\varepsilon) = 0$ and $u_1(\varepsilon) = \varepsilon$ since $v(\psi,\mathbf{x})$ itself is the potential function in the solution system of interest. It is then possible to show that the Kirkwood charging formula is given by

$$\Delta\mu = \int_0^1 d\lambda \int d\varepsilon \frac{\partial u_\lambda(\varepsilon)}{\partial\lambda} \rho^e(\varepsilon;u_\lambda) \tag{6-2}$$

where $\rho^e(\varepsilon;u_\lambda)$ is the ensemble average of Eq. (6-1) in the presence of the solute-solvent interaction u_λ. The superscript e is attached to mean the representation over the energy coordinate ε. The Legendre transform is also possible as

$$\Delta\mu = \int d\varepsilon u_1(\varepsilon)\rho^e(\varepsilon;v) - \int_0^1 d\lambda \int d\varepsilon u_\lambda(\varepsilon)\frac{\partial\rho^e(\varepsilon;u_\lambda)}{\partial\lambda} \equiv \int d\varepsilon\varepsilon\rho^e(\varepsilon) - F^e\left[\rho^e(\varepsilon)\right] \tag{6-3}$$

where $\rho^e(\varepsilon;v)$ at the potential v of interest is denotes as $\rho^e(\varepsilon)$ for notational brevity. The second term of the right hand of Eq. (6-3) is the density-functional in the energy representation. The first term is equal to the average sum of the solute-solvent interaction energy. $\Delta\mu$ and F^f are related to each other with Legendre transform since the map is proved to be one-to-one from the solute-solvent interaction potential to the distribution function [20]. the indirect part ω^e of the potential of mean force in the energy representation can be introduced correspondingly simply by rewriting $\mathbf{x}$ of Eq. (4-4) with ε as

$$\rho_\lambda^e(\varepsilon) = \rho_0^e(\varepsilon)\exp\left(-\beta\left(u\left(\varepsilon;\rho_\lambda^e\right) + \omega^e\left(\varepsilon;\rho_\lambda^e\right)\right)\right) \tag{6-4}$$

where $\rho^e{}_\lambda(\varepsilon)$ is the distribution function at the coupling parameter λ and β is the inverse of the product of the Boltzmann constant k_B and the temperature T. The density-functional is then expressed exactly as

$$F^e\left[\rho^e(\varepsilon)\right] = k_B T \int d\varepsilon \begin{bmatrix} \left(\rho^e(\varepsilon) - \rho_0^e(\varepsilon)\right) - \rho^e(\varepsilon)\log\left(\dfrac{\rho^e(\varepsilon)}{\rho_0^e(\varepsilon)}\right) \\ -\beta\left(\rho^e(\varepsilon) - \rho_0^e(\varepsilon)\right)\displaystyle\int_0^1 d\lambda\omega^e\left(\varepsilon;\rho_\lambda^e\right) \end{bmatrix} \tag{6-5}$$

when u_λ is taken so that $\rho^e{}_\lambda$ varies linearly against λ. Note the parallelism of Eqs. (6-2)-(6-5) to Eqs. (4-2)-(4-5). In Eqs. (6-4) and (6-5), u and ω are written to depend on the distribution function $\rho^e{}_\lambda$, instead of the potential u_λ, by virtue of the property of

one-to-one correspondence [20]. Equation (6-5) is an exact expression. An approximation is introduced into the λ integral of ω^e in Eq. (6-5). In the energy representation, the HNC-type and PY-type approximations are obtained by assuming the linear dependencies of ω^e and exp $(-\beta\omega^e)$-1 on λ, respectively. Although Eq. (4-5) is hard to implement due to the high-dimensionality of $(\psi,\mathbf{x})$, Eq. (6-5) is straightforward to handle since ε is one-dimensional.

In the currently used version of the method of energy representation [21-23], the solvation free energy $\Delta\mu$ is approximately expressed in terms of distribution functions constructed from $\hat{\rho}^e$ in the solution and reference solvent systems. In our treatments, the solution system refers to the system in which the solute molecule interacts with the solvent under the solute-solvent interaction v of interest at full coupling. In the solution, the average distribution ρ^e of the v value is relevant in the approximate construction of $\Delta\mu$ and is given by

$$\rho^e(\varepsilon) = \left\langle \hat{\rho}^e(\varepsilon) \right\rangle \tag{6-6}$$

where $\langle \cdots \rangle$ represents the ensemble average in the solution system of interest. On the other hand, the reference solvent system denotes the system in which the solute does not interact with the solvent and the solvent configuration is generated only by the solvent-solvent interaction. No coupling is present between the solute and solvent configurations. At an instantaneous configuration of the reference solvent system, $\hat{\rho}^e$ is constructed by placing the solute molecule in the system as a test particle. This point is parallel to the case of the particle insertion method described in Sec. 2. A simulation of the reference solvent (pure solvent) system and a simulation of a single solute molecule are performed independently, and the solute is inserted randomly into the solvent system. In this insertion process, the configuration of the solvent molecules constituting the reference solvent system is unchanged. The pair interaction energy between solute and solvent is calculated after the solute insertion without modification of the solvent configuration, and $\hat{\rho}^e$ is obtained as the histogram of the solute-solvent pair energy. The average distribution ρ_0^e and the correlation matrix χ_0^e then appear in the approximate functional for $\Delta\mu$ and are expressed, respectively, as

$$\rho_0^e(\varepsilon) = \left\langle \hat{\rho}^e(\varepsilon) \right\rangle_0 \tag{6-7}$$

and

$$\chi_0^e(\varepsilon,\eta)=\left\langle\hat{\rho}^e(\varepsilon)\hat{\rho}^e(\eta)\right\rangle_0-\left\langle\hat{\rho}^e(\varepsilon)\right\rangle_0\left\langle\hat{\rho}^e(\eta)\right\rangle_0 \tag{6-8}$$

where $\langle\cdots\rangle_0$ represents the ensemble average in the reference solvent system. In the sampling corresponding to $\langle\cdots\rangle_0$, the solute and solvent degrees of freedom are uncoupled from each other in the probability distribution.

An approximate functional for $\Delta\mu$ is derived in Ref. [21]. The functional is constructed by adopting the PY-type approximation in the unfavorable region of the solute-solvent interaction and the HNC-type approximation in the favorable region. $\Delta\mu$ is then given by a set of definitions and equations listed as

$$w^e(\varepsilon)=-k_BT\log\left(\frac{\rho^e(\varepsilon)}{\rho_0^e(\varepsilon)}\right)-\varepsilon \tag{6-9}$$

$$w_0^e(\varepsilon)=-k_BT\int d\eta\left(\frac{\delta(\varepsilon-\eta)}{\rho_0^e(\varepsilon)}-\left(\chi_0^e\right)^{-1}(\varepsilon,\eta)\right)\left(\rho^e(\eta)-\rho_0^e(\eta)\right) \tag{6-10}$$

$$\begin{aligned}\Delta\mu=\int d\varepsilon\rho^e(\varepsilon)-k_BT\int d\varepsilon\Big[&\left(\rho^e(\varepsilon)-\rho_0^e(\varepsilon)\right)-\rho^e(\varepsilon)\log\left(\frac{\rho^e(\varepsilon)}{\rho_0^e(\varepsilon)}\right)\\&-\left\{\alpha(\varepsilon)F(\varepsilon)+(1-\alpha(\varepsilon))F_0(\varepsilon)\right\}\left(\rho^e(\varepsilon)-\rho_0^e(\varepsilon)\right)\Big]\end{aligned} \tag{6-11}$$

$$F(\varepsilon)=\begin{cases}\beta w^e(\varepsilon)+1+\dfrac{\beta w^e(\varepsilon)}{\exp\left(-\beta w^e(\varepsilon)\right)-1} & (\text{when } w^e(\varepsilon)\le 0)\\ \dfrac{1}{2}\beta w^e(\varepsilon) & (\text{when } w^e(\varepsilon)\ge 0)\end{cases} \tag{6-12}$$

$$F_0(\varepsilon)=\begin{cases}-\log\left(1-\beta w_0^e(\varepsilon)\right)+1+\dfrac{\log\left(1-\beta w_0^e(\varepsilon)\right)}{\beta w_0^e(\varepsilon)} & (\text{when } w_0^e(\varepsilon)\le 0)\\ \dfrac{1}{2}\beta w_0^e(\varepsilon) & (\text{when } w_0^e(\varepsilon)\ge 0)\end{cases} \tag{6-13}$$

$$\alpha(\varepsilon)=\begin{cases}1 & (\text{when } \rho^e(\varepsilon)\ge\rho_0^e(\varepsilon))\\ 1-\left(\dfrac{\rho^e(\varepsilon)-\rho_0^e(\varepsilon)}{\rho^e(\varepsilon)+\rho_0^e(\varepsilon)}\right)^2 & (\text{when } \rho^e(\varepsilon)\le\rho_0^e(\varepsilon))\end{cases} \tag{6-14}$$

When Eqs. (6-9)-(6-14) are used to evaluate the solvation free energy, the inputs are the three energy distribution functions ρ^e, ρ_0^e, χ_0^e given by Eqs. (6-6)-(6-88),

respectively. ρ^e is obtained from a simulation of the solution system of interest, and ρ_0^e and χ_0^e are constructed with a simulation of the reference solvent system. The approximate scheme provided by Eqs. (6-6)-(6-14) does not require simulations at the intermediate states of solute insertion. This leads to substantial reduction in free-energy calculation. On the other hand, when Eqs. (6-9)-(6-14) are used with ρ^e, ρ_0^e, χ_0^e obtained from molecular simulation, a possible artifact in simulation arising, for example, from the use of force field, periodic boundary condition and/or the handling scheme of electrostatic interaction (cutoff, Ewald, particle-mesh Ewald, and so on), is generally retained.

It should be noted that when the solute molecule is inserted into the reference solvent system, it often overlaps with solvent molecules. The overlapping configurations contribute to ρ_0^e and χ_0^e at large energy coordinates and accounts for the excluded volume effect in the solvation free energy. In addition, Eqs. (6-9)-(6-14) show that the ε value itself acts only as an index for the distribution functions ρ_0^e and χ_0^e when ε is large enough and the corresponding ρ^e is (essentially) zero. The ε value in the excluded volume region is not used in the calculation of $\Delta\mu$, except in the construction of ρ_0^e and χ_0^e; $\Delta\mu$ calculation in the large ε region is not affected by the change in the ε value as far as ε varied corresponds to the excluded volume. This property is of course satisfied for the exact functional for the solvation free energy, and is kept valid for the approximate one given by Eqs. (6-9)-(6-14).

To see the origin of the above property, we consider a family of solute-solvent interactions $u(\varepsilon)$, as done around Eq. (6-1), which are constant over an equienergy surface of the potential $v(\psi,\mathbf{x})$ of interest. It is possible to show through a line parallel to the blip-function method [1] that

$$\frac{\delta(\Delta\mu)}{\delta\exp(-\beta u(\varepsilon))} = -k_B T \exp(\beta u(\varepsilon))\rho^e(\varepsilon;u) \tag{6-15}$$

where $\rho^e(\varepsilon;u)$ is the ensemble average of Eq. (6-1) in the presence of the solute-solvent interaction $u(\varepsilon)$. Here we restrict our attention only to the ε region corresponding to the

excluded volume and the set of $u(\varepsilon)$ which are prohibitively large in that ε region. Then, the Boltzmann factor $\exp(-\beta u(\varepsilon))$ is exponentially small in this region. Further, for a change of $u(\varepsilon)$ within the set, the corresponding variation of $\exp(-\beta u(\varepsilon))$ is also vanishingly small. It then follows that such variations of $u(\varepsilon)$ do not change the $\Delta\mu$ value. This is shown by the fact that the right-hand side of Eq. (6-15) remains to be finite even when $u(\varepsilon)$ is large as is generally proven in Ref. [1]. Thus, the choice of the $u(\varepsilon)$ form has no effect on the $\Delta\mu$ calculation in the excluded-volume region.

In Figure 4, the approximate values of the solvation free energy $\Delta\mu$ for typical solute molecules in solvent water are compared to the corresponding exact values obtained from the free-energy perturbation method. The (solvent) density of 1.0 g/cm^3 and the temperature of 25 ºC is an ambient state, and the densities of 1.0, 0.6, and 0.2 g/cm^3 at 400 ºC correspond to high-, medium-, and low-density supercritical states. The good agreement is observed between the approximate and exact values. The agreement is particularly notable at the medium- and low-density states of 0.6 and 0.2 g/cm^3 and 400 ºC. When the solute is ionic, the density at the state of 0.2 g/cm^3 and 400 ºC is not yet low in the sense, for example, that the hydration number at that state is comparable to the numbers at ambient states [24,25]. Even in this case, our approximate procedure is effective in determining the solvation free energy. The solvation free energies of water at 1.0 g/cm^3 and 400 ºC and of methanol and ethanol at 0.6 g/cm^3 and 400 ºC are rather small in magnitude. These behaviors are caused by the balance between the favorable and unfavorable contributions of the solute-solvent interactions, and are well reproduced by our approximate method. Therefore, the single functional given by Eqs. (6-6)-(6-14) provides an accurate and efficient route to the solvation free energy for various types of solutes over a wide range of thermodynamic conditions.

By virtue of Eq. (4-1), the average sum $\langle u \rangle$ of the solute-solvent interaction energy in the solution system of interest is smaller than or equal to $\Delta\mu$. This means that the density-functional F^e is always non-positive for any solute-solvent distribution function. Actually, the density functional is a measure of the "difference" between ρ^e in solution and ρ_0^e in the reference solvent. It is zero only when $\rho^e = \rho_0^e$. The density-functional is expected to be more negative when ρ^e and ρ_0^e appear more differently. On the other hand, the first term of Eq. (6-3) is equal to $\langle u \rangle$, and is more negative when ρ^e is more populated in the low-energy region of ε. A typical behavior is that ρ_0^e reduces monotonically toward the low-energy tail. Thus, the first term of Eq. (6-3) is more

negative when ρ^e and ρ_0^e are more different. This indicates that the first and second terms of Eq. (6-3) fluctuate to the same direction through the variation of ρ^e. It is then expected that $\Delta\mu$ of Eq. (6-3) converges faster in molecular simulation than its components expressed as the first and second terms of Eq. (6-3). Indeed, usual experience is that when (an approximate form of) Eq. (6-5) is employed, the solvation free energy $\Delta\mu$ converges faster than the average sum $\langle u \rangle$ of the solute-solvent interaction energy in solution.

The convergence behavior described above is also seen from the variation principle [26]. According to the density-functional theory in the energy representation [20,22],

$$\frac{\delta(\Delta\mu)}{\delta\rho^e(\varepsilon)} = 0 \tag{6-16}$$

holds when ρ^e is equal to the equilibrium function distribution ρ_{eq}^e. Equations (6-3) and (6-16) then show that when ρ^e varies around ρ_{eq}^e, the change in the solute-solvent interaction energy $\langle u \rangle$ is compensated by the change in the density-functional F^e. In other words, $\langle u \rangle$ and F^e vary into the same direction. The $\Delta\mu$ variation is thus smaller since $\Delta\mu$ is given by the difference between $\langle u \rangle$ and F^e. Actually, the equilibrium distribution ρ_{eq}^e is the one obtained from molecular simulation performed to convergence. ρ_{eq}^e from the simulation satisfies Eq. (6.16) for the exact form of F^e. In practice, an approximate form of F^e is employed as listed by Eqs. (6-9)-(6-14), and Eq. (6.16) is not necessarily satisfied for ρ_{eq}^e obtained from the converged simulation. The compensation between the $\langle u \rangle$ and F^e fluctuations is effective, nevertheless, when the approximation is good.

In the energy representation, the interaction energy between the solute and solvent is adopted as the one-dimensional coordinate of the distribution functions, and a functional for the solvation free energy is constructed from energy distribution functions in the solution and reference solvent systems of interest. The introduction of the energy coordinate for distribution functions is a kind of coarse-graining procedure for reducing the information content of the solute-solvent configuration. Since the set of configurations (structures) with the equal solute-solvent interaction energies are grouped into a unit in the energy representation, any approximate functional built in

terms of energy distribution functions cannot violate the statistical-mechanical principle that the configurations with the same solute-solvent interaction energies contribute to the solvation free energy with the equal weights.

As seen in Eq. (6-1), each of the solute and solvent molecules is taken as a single unit in the energy representation. The molecule is treated as a whole, while the coordinate for the distribution functions is one-dimensional. No explicit reference is made to the detail of the molecular structure by focusing on the interaction energy. The following advantages then emerge in the method of energy representation.

Firstly, the method is straightforwardly applicable to molecules with intramolecular flexibility. The implementation is indifferent whether the molecule is rigid or flexible. The information of structural fluctuation of the molecule is adsorbed when the energy coordinate is introduced by Eq. (6-1). For large molecules constituting micellar, membrane, and protein systems, it is not allowed to neglect the molecular flexibility. In the method of energy representation, an additional and/or separate scheme is not necessary to be formulated for large, flexible species.

Secondly, the treatment of inhomogeneous system and clusters is straightforward. So far, the formulation does not assume the system homogeneity and the thermodynamic limit. The application to inhomogeneous and/or finite systems is then possible without modification. The binding of a molecule to such nanoscale structures as micelle, membrane, and protein can be viewed as a solvation in an inhomogeneous and finite, mixed solvent [26,27]. The method of energy representation can thus be a useful approach to intermolecular correlation and association important in biological and interface sciences. Example applications to micellar and membrane systems are presented in Sec. 7. In connection to the inhomogeneous system, the application is straightforward to a system with external field. Actually, the treatment of the external field is mathematically identical to the treatment of the intramolecular degrees of freedom (molecular flexibility). This is possible since both of them are usually expressed as one-body potentials. The difference arises from the functional form of the potential, most notably the range of interaction, which is the origin of diverse physical effects. The mathematical development can be identical, however, since the formulation does not refer explicitly to the form of potential function.

Thirdly, an accurate treatment is possible for supercritical fluid. In supercritical fluid, the solvent density and temperature can varied over wide range and the solvent effect may act as a key to control a chemical process. It is well-known that supercritical fluid can be described accurately when the whole molecule is treated as a single unit [1]. A multidimensional representation is necessary, however, in the usual coordinate space.

By introducing the energy as the coordinate for distribution functions, the whole molecule can be taken as a single unit with keeping the description one-dimensional. The approximate functional given by Eqs. (6-6)-(6-14) incorporates the intermolecular correlation at the two-body level. The solvation free energy obtained is then exact to second order in the solvent density. Since the method is exact in the low-density regime, a formulation of a good approximation in the high-density regime leads to an accurate description over a wide range of solvent density.

Finally, the combination with the quantum-mechanical/molecular-mechanical (QM/MM) methodology can be performed. The QM/MM method treats the solute (reactive species) with a quantum-chemical scheme and the environment (solvent) with a classical force field. It is one of the frontline methods in modern theoretical/computational chemistry. In QM/MM calculation, the many-body effect is introduced for the solute-solvent interaction and is beyond the applicability of conventional theories of solutions. In the method of energy representation, the fluctuation of the electronic state in response to the environment is viewed as a fluctuation of intramolecular degrees of freedom of the QM solute. The evaluation becomes feasible for the free energy for the many-body effect of the electronic fluctuation. In addition, Eq. (6-1) makes no reference to the functional form of the potential function. It refers only to the value of the potential energy, and there is no need for deterioration or modification of the electronic-state calculation. Thus, the treatment is possible for an arbitrary distribution of charges. The contraction to a set of point charges is not necessary, and the effect of the diffuse (cloud-like) nature of electronic distribution can now be determined. The detail of the combination with the QM/MM methodology is given in Ref. [23]. The addition of an electron (reduction) can also be treated as a solvation in an inhomogeneous mixed-solvent system. By employing the argument of inhomogeneous systems described above, the added electron is viewed as the solute and the oxidized form and the solvent (in the usual sense) as the mixed solvent. With this view, the computation of the reduction potential in condensed phase is now possible with the QM/MM calculation [28].

7 Application to molecular binding into micelle and lipid-membrane systems

Micelle is a self-assembled aggregate of amphiphilic surfactants in water. It involves a hydrophobic core and provides a favorable environment for organic compounds. Upon formation of a micelle, the solubility is often enhanced for an organic compound which is insoluble or sparingly soluble in water. This phenomenon is called solubilization, and is a most important function of a micelle.

Membrane is a soft, self-organizing aggregate of amphiphilic lipids in aqueous solution. It distinguishes one side of the solution from the other, and plays important roles in distribution and transport of a molecule. The key process toward the membrane function is the binding of a molecule into membrane.

On the microscopic level, the solubilization and membrane-binding are common in that they correspond to the transfer of a molecule from bulk water to the inside (or surface) of a nanoscale, self-organizing structure in solution. In both cases, the process is quantified by the (standard) free energy of binding. The binding free energy is determined by the cooperation and/or competition of the interactions among the surfactant or lipid, the solute molecule to be bound, and the solvent (water and cosolvents if present). A unified framework for the free-energy analysis at atomic resolution is thus desirable to be developed for the molecular understanding and control of the binding in micelle and membrane.

The basic idea of our scheme is to view a micellar or membrane solution as a mixed solvent. The surfactant or lipid molecules are treated not as solute species, but as part of the mixed solvent system. The molecule to be bound into the micelle or membrane is the only species regarded as the solute. In this view, the solvation free energy of the solute molecule in a micellar or membrane solution denotes the free-energy change for turning on the interactions of the solute with the solvent water, surfactants or lipids, and counterions if present. The binding is described by the difference in the solvation free energy between the micellar or membrane system and the neat solvent (pure water) system. When the host (micelle or membrane) is dilute, in particular, it is natural to set the origin of the solution at (or near) the center of the host. The mixed solvent system then consists of the surfactant or lipid molecules forming the host, water, and distributed counterions (when the surfactant or lipid is ionic). Water is deficient in the interior of the host, and the surfactant or lipid molecules belonging to the host are localized around the origin. The mixed solvent is thus inhomogeneous even before the solute insertion. When the host is micelle, furthermore, the number of surfactants is finite since the aggregation number is typically 10^2-10^4. The micellar solution is partially finite in the sense that the number is finite for the surfactant (and counterion) involved as solvent species in the dilute micellar solution. The point of our scheme is to evaluate the solvation free energy of a solute in an inhomogeneous and partially finite, mixed solvent system.

Even when the system is inhomogeneous and partially finite, a formally exact free-energy calculation is possible by the standard free energy perturbation and thermodynamic integration methods [1-3]. These methods are notorious for the heavy

computational demand, however, since an explicit reference needs to be made to the intermediate states of the gradual variation process of the system. We combine the large-scale molecular simulation with the method of energy representation to evaluate the binding free energy.

The surfactant employed is sodium dodecyl sulfate (SDS) and the lipid is 1,2-dimyristoyl-*sn*-glycero-3-phosphatidylcholine (DMPC). Their structures are illustrated in Figure 5. The solute molecule is methane, benzene, and ethylbenzene in the micelle simulation, and is CO, benzene, and ethylbenzene in the membrane simulation. The computational procedures are fully described in Refs. [26,27].

In Figure 6, we show the density profile of SDS micelle and DMPC bilayer. Figure 6 (a) gives the densities of the hydrophobic tail, the headgroup, and water as functions of the distance r from the micellar center identified as the center of mass of the dodecyl sulfate anions. Since the eccentricity of the SDS micelle is small, it is natural to divide the micellar system into a set of regions on the basis of Figure 6 (a). We introduce 6 regions in terms of the center-of-mass distance r of the solute from the micellar center by concentrically dividing the domain of $r < 30$ Å with an equal interval of 5 Å. In the following, the regions are numbered I … VI from the micelle inside to outside. According to Figure 6 (a), the regions I-III corresponds to the hydrophobic core, where water is scarcely present. The region IV refers to the headgroup region in contact with water, and is the transition region from the hydrophobic core to the aqueous region outside the micelle represented by the regions V and VI.

In the membrane system simulated, the z-axis is taken to be normal to the plane of the DMPC bilayer and $z = 0$ is set to the z-coordinate of the center of mass of the DMPC molecules. In Figure 6 (b), we show the densities of the hydrophobic tail, the glycerol backbone, the hydrophilic headgroup, and water obtained from the simulation of DMPC and water without solute. Since the (converged) distribution of particles is symmetric against the reflection at $z = 0$, the density averaged over the positive and negative z–domains is displayed only against the positive z -abscissa. As done for the SDS micelle, we introduce 6 regions of z by dividing the domain of $|z| < 30$ Å with an equal interval of 5 Å. The regions are numbered I … VI from the membrane inside to outside. According to Figure 6 (b), the regions I and II correspond to the hydrophobic interior, where water is scarcely present. The glycerol and headgroup are mainly located in the regions III-V, where the water density gradually varies. The density profile recovers the bulk behavior in the region VI, and the thickness of (one leaflet of) the DMPC layer is ~20 Å.

In Figure 7, we show the solvation free energy $\Delta\mu$ of the solute in the regions I-VI

and in the bulk water. The bulk denotes the region far from the host (micelle or membrane), and $\Delta\mu$ in the bulk is the solvation free energy in neat water. Evidently, each hydrophobic solute is free-energetically stabilized in the inside of the host.

When the host is micelle, the distribution within the hydrophobic interior (regions I-III) is rather diffuse. For benzene and ethylbenzene in the micelle, the $\Delta\mu$ difference between the regions IV and I-III shows that these solutes are localized in the hydrophobic core of the micelle. When the solute is methane and is bound into the SDS micelle, in contrast, the $\Delta\mu$ difference is small and the probability of finding the solute is appreciable also in the headgroup region. Thus, the solubilization of benzene and ethylbenzene is more sharply characterized as a transfer from the aqueous to hydrophobic environment. For all the solutes examined, the distribution is negligible in the aqueous regions V-VI outside the micelle. This is a support to the pseudophase model, and the support is stronger for benzene and ethylbenzene.

When the host is membrane, Figure 7 (b) shows that $\Delta\mu$ varies by $\sim k_BT$ within the regions I-IV. A strong tendency of localization is not observed, and the hydrophilic portion of DMPC also interacts with the hydrophobic solute. In the SDS micelle, benzene and ethylbenzene are more strongly localized in the hydrophobic interior. This contrast corresponds, as evidenced in the molecular structure of Figure 5, to the fact that DMPC involves methylene and methyl groups in the hydrophilic segment unlike SDS.

8 Perspectives

In the present review, we introduce a new theory of solutions. The theory is formulated to satisfy the requirements for applications to diverse systems listed as

(a) flexible molecule (and ion)
(b) inhomogeneous system
(c) supercritical fluid covering wide range of density from gas-like, low density to liquid-like, high density
(d) QM/MM system involving many-body interactions

The treatment of these systems becomes possible by adopting the solute-solvent interaction energy as the coordinate of distribution functions and formulating the density-functional theory over the energy coordinate. The distribution functions are obtained from molecular simulation, and an efficient and accurate free-energy calculation is now possible with an (approximate) functional.

According to Eqs. (6.9)-(6.14), a numerical difficulty arises when ρ^e is non-zero and ρ_0^ε is zero in some region of ε (there is no problem, on the other hand, when ρ^e is zero). This is encountered when the solute or solvent changes its (intramolecular) structure

significantly through solute-solvent interaction. When the variation of the solute structure is large, the difficulty can actually be circumvented by modifying the sampling scheme with knowledge of computational results in solution [22]. The functional for the solvation free energy is then modified in response to the sampling scheme. A challenge is present when the (intramolecular) structure changes drastically for the solvent. When the protein and water are viewed as a mixed solvent, the induced fit corresponds to such category.

The concept of energy coordinate greatly expands the applicability of solution theory. A particular example is the combination with the QM/MM method. In the QM/MM method, the solute-solvent interaction is not pairwise additive any more and is beyond the applicability of conventional theory of solutions. With the energy concept, furthermore, micelle and membrane are now within the unified scope of the solution theory. The extensions to protein system and the connection to coarse-graining description are awaited.

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Figures and captions

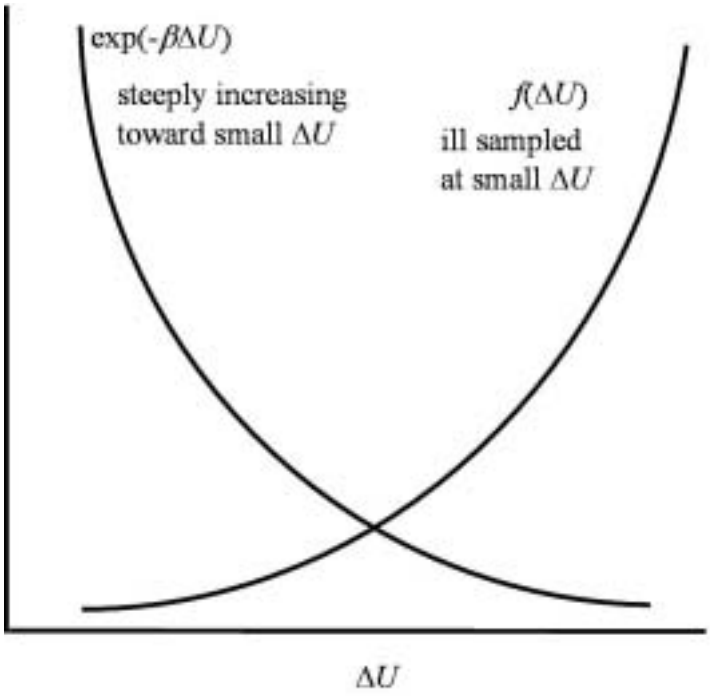

Figure 1. A schematic distribution of ΔU value in the particle insertion method.

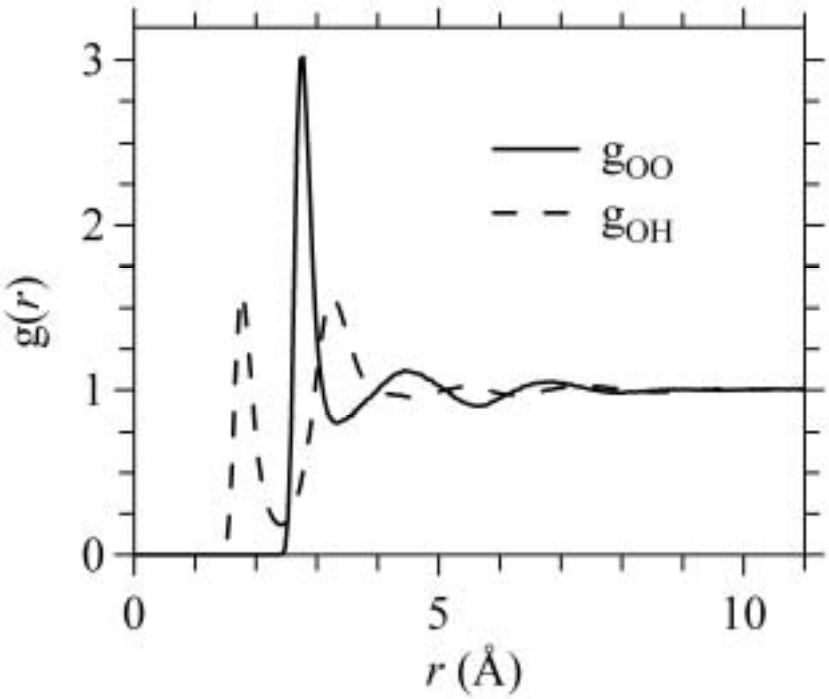

Figure 2. O-O and O-H radial distribution functions of water at 1 g/cm^3 and 25 °C.

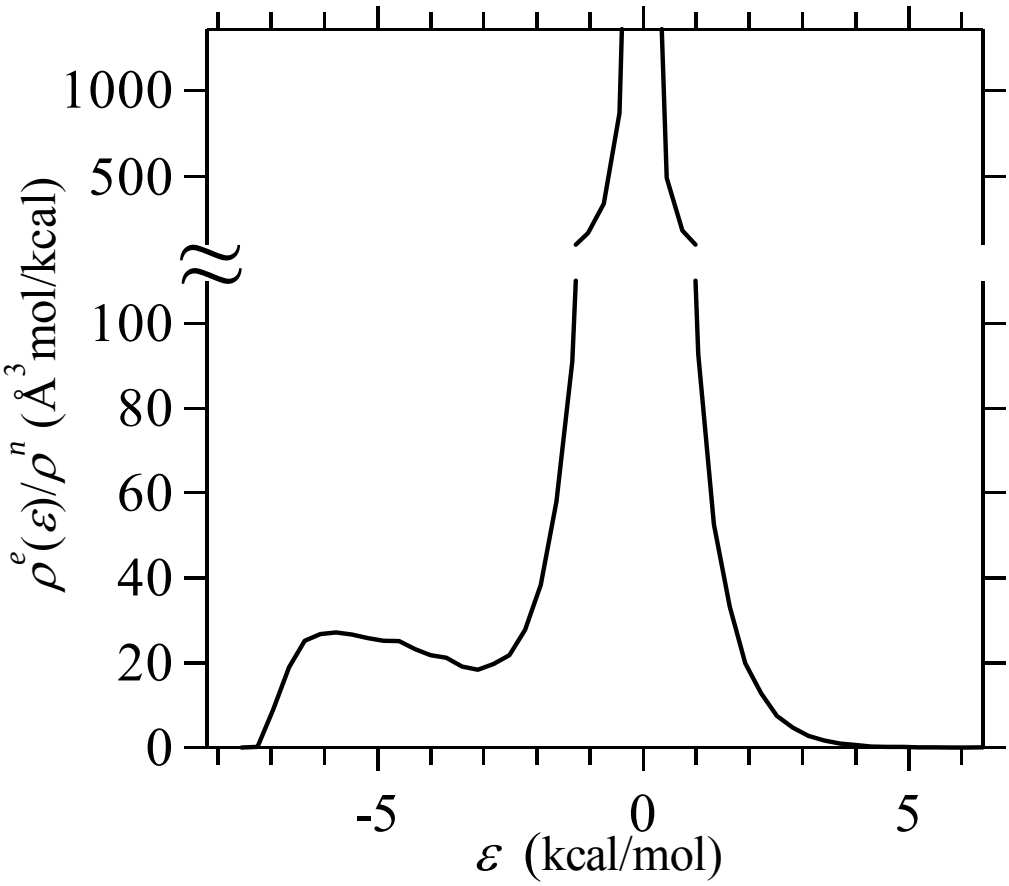

Figure 3. The distribution function of the pair interaction energy ε of water at 1 g/cm^3 and 25 °C. The graduation in the ordinate is changed at 100 Å^3 mol/kcal.

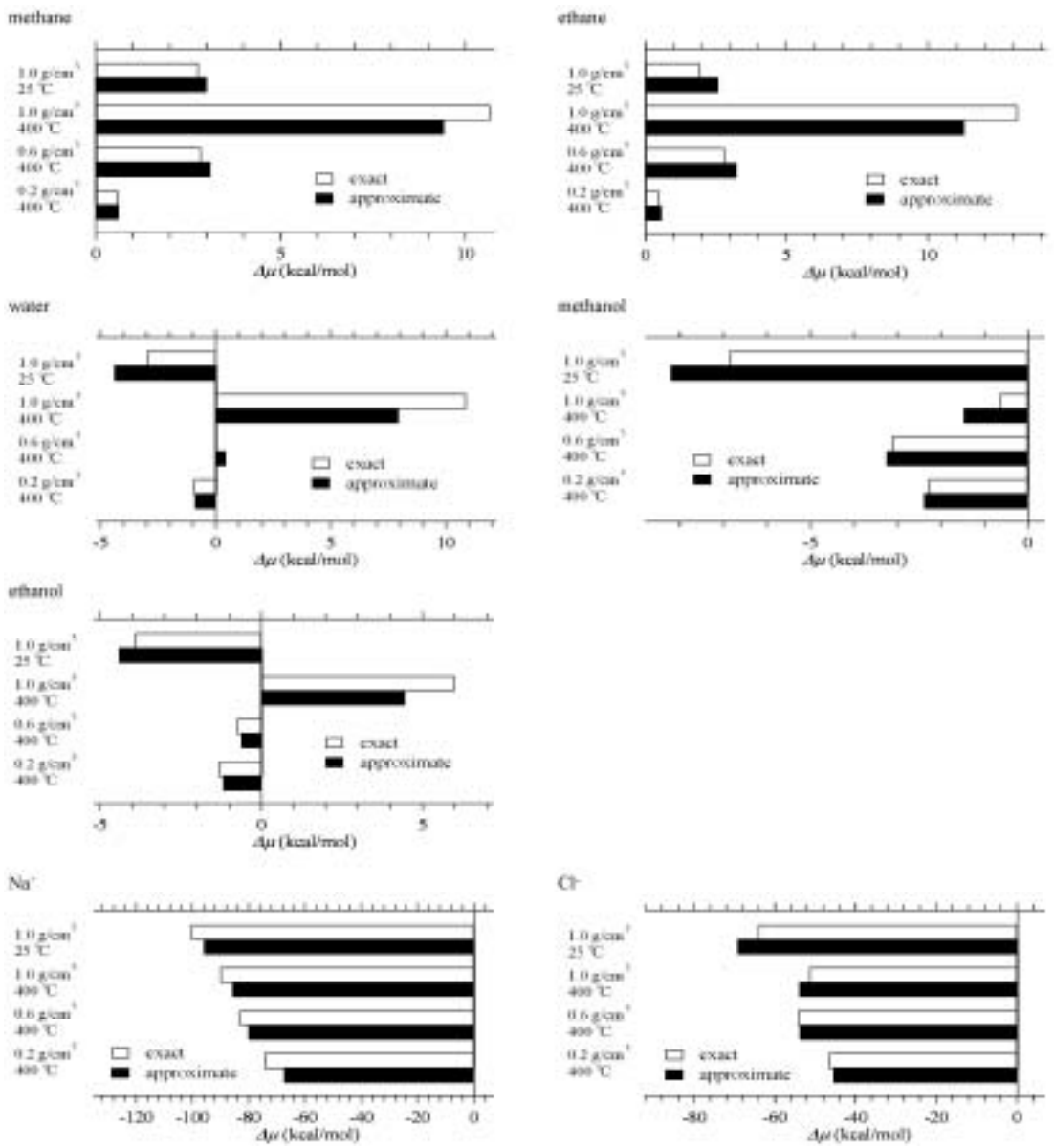

Figure 4. The solvation free energy $\Delta\mu$ calculated by the approximate method of energy representation and the exact, free-energy perturbation method. The thermodynamic state is specified by the solvent density and temperature.

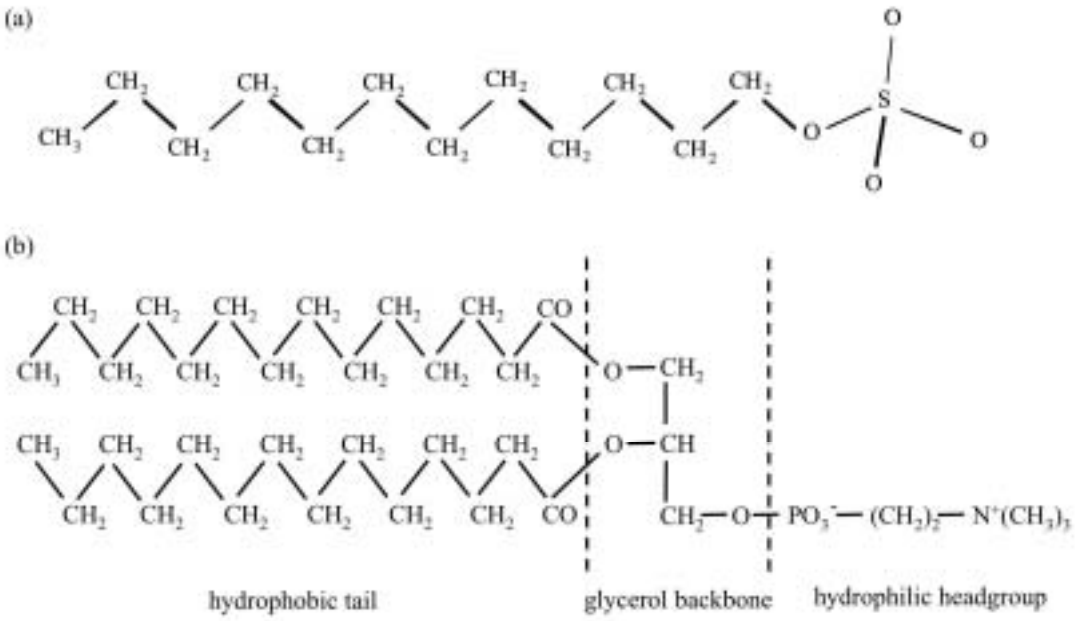

Figure 5. (a) The structure of the dodecyl sulfate anion. (b) The structure of DMPC. The molecule is divided into three portions of the hydrophobic tail, glycerol backbone, and hydrophilic headgroup. In the present work, the division is done as denoted in the figure, though the diving scheme is always of some ambiguity.

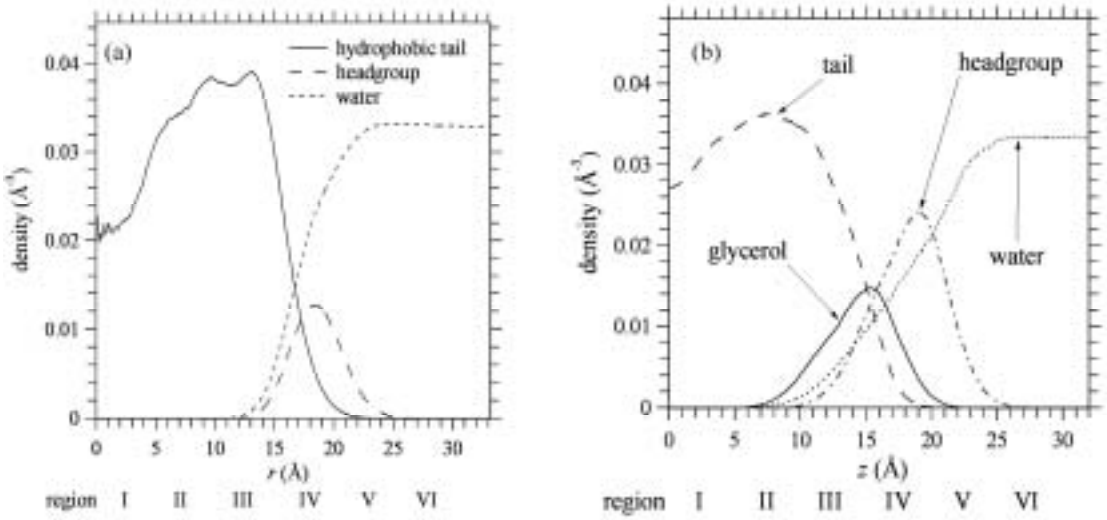

Figure 6. (a) The densities of the hydrophobic tail, the headgroup, and water in the SDS micellar system as functions of the distance r from the center of mass of the dodecyl sulfate anions forming the micelle. The density of the hydrophobic tail refers to the sum of the (number) densities of the methyl and methylene groups of the dodecyl sulfate anion, and the density of the headgroup is the sum (number) density of the sulfur and oxygen atoms. The water density is expressed with respect to the center of mass of the water molecule. Six regions are introduced by dividing the domain of $r < 30$ Å with an interval of 5 Å, and are numbered I … VI from the micelle inside to outside. (b) The densities of the hydrophobic tail, the glycerol backbone, the hydrophilic headgroup, and water in the DMPC membrane system as functions of the separation z along the z-axis from the center of mass of the DMPC molecules. The densities obtained over the positive and negative z-domains are averaged, and the averaged density is shown against the positive z-abscissa. The densities of the tail and glycerol refer to the sums of the (number) densities of the carbons and oxygens contained in the tail and glycerol portions of Figure 5 (b), respectively, and the density of the headgroup is the sum of the (number) densities of the carbons, oxygens, nitrogen, and phosphorus in the corresponding portion. The water density is expressed with respect to the center of mass of the water molecule. Six regions are introduced by dividing the domain of $z < 30$ Å with an interval of 5 Å, and are numbered I … VI from the membrane inside to outside.

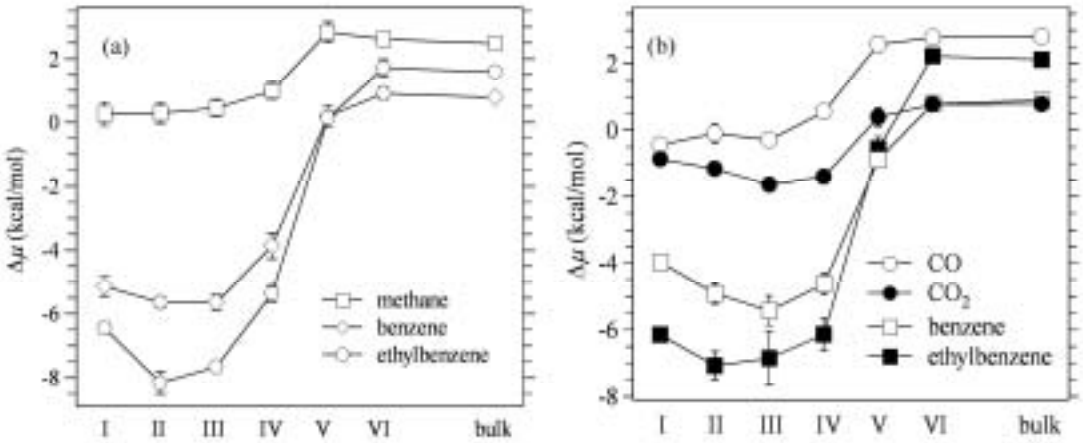

Figure 7. The solvation free energy $\Delta\mu$ in the regions I-VI and in the bulk for (a) the SDS micellar and (b) the DMPC membrane systems. The bulk denotes the region far from the host, and $\Delta\mu$ in the bulk is equal to the solvation free energy of the solute inserted into neat water. The error bar is expressed at 95% confidence level, and is smaller than the size of the corresponding data symbol when it is not shown. The lines connecting the data are drawn for eye guide.